

University of Bahrain

College of Information Technology
Department of Computer Science

ITCS347 Design and Analysis of Algorithms

Exercises #3– **Brute Force**

1. Given an array of integers A of size n and a target sum S , find all unique pairs (i, j) , where $i \neq j$, such that $A[i] + A[j] = S$.
2. Given an array A of integers, find the *maximum sum* of any contiguous subarray.
3. Given a string S , find the length of the *longest substring* without repeating characters.
4. Given a string, S , of n digits in the range from 0 to 9, describe a brute-force algorithm for converting S into the integer it represents. What is the running time of your algorithm?
5. Write a brute-force algorithm that removes all the duplications from an array given as input parameter. What is the running time of your algorithm?
6. The greatest common divisor between any two positive integers a and b can be written as a linear combination between these two integers. I mean, there exist integers s and t such that $\gcd(a, b) = sa + tb$. Write a brute-force algorithm that computes $\gcd(a, b)$ and finds the values s and t , for a given positive integers a and b .
7. Suppose you are given an array, A , of n positive integers. Describe an $O(n)$ brute-force algorithm for removing all the even numbers from A . That is, if A has m odd numbers, then, after you are done, these odd numbers should occupy the first m cells of A in the same relative order they were in originally.
8. Given a 9×9 **Sudoku grid**, determine whether it is a valid solution according to the rules of Sudoku. A valid Sudoku must satisfy the following conditions:
 1. Each row must contain all digits from 1 to 9 without repetition.
 2. Each column must contain all digits from 1 to 9 without repetition.
 3. Each of the nine 3×3 **subgrids** (also called **boxes**) must contain all digits from 1 to 9 without repetition.

The input to the algorithm is a 2D array $grid[9, 9]$ where $grid[i, j]$ represents the value at row i , column j . Empty cells may be represented by '0' or '.', but in a solved Sudoku, all cells must be filled with digits 1 – 9.

9. Given two rectangles in a 2D plane, each defined by their bottom-left and top-right coordinates, determine whether they overlap (have any area in common). Each rectangle is represented by two points: Bottom-left corner (x_1, y_1) and Top-right corner (x_2, y_2) . Two rectangles R_1 and R_2 overlap if there exists at least one point (x, y) that lies within both rectangles.

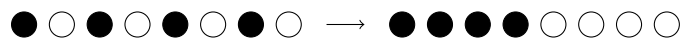
Example.

Rectangle R_1 : (1, 1) to (4, 4)

Rectangle R_2 : (2, 2) to (5, 5)

These rectangles overlap because they share the region from (2, 2) to (4, 4).

10. You have a row of $2n$ disks of two colors, n dark and n light. They alternate: dark, light, dark, light, and so on. You want to get all the dark disks to the right-hand end, and the light disks to the left-hand end. The only moves you are allowed to make are those that interchange the positions of two neighboring disks. Design a brute force algorithm for solving this puzzle and determine the number of moves it takes.



11. A warehouse has n products, each with a demand $d[i]$ and storage cost $s[i]$. It is required to find which products to restock that maximizes profit without exceeding capacity C . What is the name of the computer science algorithm that can be used to solve this problem? Apply the algorithm on $d = \{10, 40, 30, 50\}$, $s = \{5, 4, 6, 3\}$, and $C = 10$.